ANNIKET PRAJAPATI

Software Engineer — VR / Real-Time Systems & Simulation Specialist

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Hiring Team — Teleoperation Systems
ALTEN Technology USA

APPLICATION: Software Engineer - Teleoperation Systems (Greenhouse ID: 5272024007)

Dear Hiring Team,

I am writing to express my enthusiastic interest in the **Software Engineer – Teleoperation Systems** role at ALTEN Technology USA. With a B.Tech in Computer Science and over 50 production real-time interactive systems engineered—including 35+ high-performance VR applications maintained at locked 90+ FPS—I specialize in building low-latency spatial control loops, 6DoF operator input pipelines, and physics-accurate simulation environments.

Your mission to pioneer intuitive, ultra-low-latency human-in-the-loop interfaces for robotic platforms aligns precisely with my core technical expertise across real-time systems, kinematics, and immersive controls:

- **Immersive 6DoF Teleoperation & Operator UX:** Architected spatial control architectures utilizing 6DoF hand tracking, spatial controllers, and custom haptic feedback mechanisms. Engineered robust translation layers converting operator positional/rotational telemetry into real-time mechanical actuators while mitigating latency jitter through predictive interpolation and dead-reckoning filters.
- **Real-Time Simulation & Testing Pipelines:** Extensive background developing physics-driven simulation testbeds in Unity and C++, including asynchronous networking pipelines (WebSockets, TCP/UDP sockets, REST architectures). Successfully integrated external scientific telemetry (such as live orbital orbital data streams via NASA NeoWs REST API) into high-fidelity dynamic physics simulations.
- **Ultra-Low-Latency Real-Time Optimization:** Deep proficiency profiling CPU/GPU performance (draw call batching, memory footprint caching, lock-free concurrency, garbage-free update cycles) to ensure determinism and zero-frame-drop throughput essential for teleoperation safety and situational awareness.
- **Cross-Disciplinary Engineering & C++/C# Mastery:** Strong foundational computer science discipline in data structures, 3D math (quaternions, coordinate space transformations, inverse kinematics), and modular middleware interfaces ready to interface with ROS/ROS2 nodes and robotic manipulators.

I am eager to contribute my deep experience in VR spatial telepresence, real-time networking, and simulation modeling to ALTEN's world-class robotics engineering initiatives. Thank you for your consideration, and I look forward to discussing how my capabilities can accelerate your teleoperation platform development.

Sincerely,

Anniket Prajapati

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